Smart Bike Rental System - Setup & Compatibility Guide

> **Purpose:** Ensure all team members use compatible versions and follow the same setup to avoid "works on my machine" issues.

Project Architecture

```
                    User (Desktop / Mobile)

                            |

        HTML + CSS + JavaScript (Responsive UI)

                            |
                     REST API Requests

                            |

               Java Spring Boot Backend

                            |

                    H2 / MySQL Database
```

Recommended Versions

Java

Component	Version
JDK	**21 LTS (Recommended)**
Minimum Supported	21
Current Local Installation	Java 25 LTS (Can be used for development)

Why Java 21?

Although Java 25 is installed locally, Java **21 LTS** has the best compatibility with:

- Spring Boot
- Maven
- Render
- Railway
- Most cloud providers

Configure Maven to compile for Java 21.

Spring Boot

Version

```
3.5.x (Latest Stable)
```

Maven

```
3.9+
```

Node.js

```
22 LTS
```

Required only if frontend build tools are added later.

Git

```
Latest Stable
```

Backend Dependencies

Include the following Spring Boot starters:

- Spring Web
- Spring Data JPA
- Spring Security
- Validation
- H2 Database
- Lombok (Optional)

Frontend Stack

- HTML5
- CSS3
- JavaScript (ES6)
- Bootstrap 5
- Chart.js
- QR Code Library

No frontend framework is required.

Maven Configuration

In `pom.xml`

```
<properties>
    <java.version>21</java.version>
</properties>
```

This ensures the project builds correctly even when developers have newer JDK versions installed.

Mobile Compatibility

The application must work on:

- Android
- iPhone
- Tablets

- Desktop
- Laptop

Requirements

- Responsive Design
- Touch-friendly UI
- Mobile Navigation
- QR Scanner Support
- Camera Access
- Geolocation Support

Browser Support

Test on

- Chrome
- Edge
- Firefox
- Safari

Deployment Architecture

Frontend

Deploy to

- GitHub Pages
- Vercel

These platforms serve only static files.

Frontend contains

```
HTML

CSS

JavaScript

Images
```

Backend

Deploy to

- Render (Recommended)
- Railway (Alternative)

Backend contains

```
Java

Spring Boot

REST APIs

Database
```

Important Deployment Note

GitHub Pages **cannot** run Java applications.

Vercel **cannot** host a long-running Spring Boot server.

Therefore, the architecture is:

```
GitHub Pages
        |

Vercel
        |

Frontend

↓

REST API

↓

Render / Railway

```

```
↓

Java Spring Boot
```

The frontend communicates with the backend using REST APIs.

Environment Variables

Backend

```
SERVER_PORT

DATABASE_URL

DATABASE_USERNAME

DATABASE_PASSWORD
```

Frontend

```
API_BASE_URL
```

Example

```
https://bike-rental-api.onrender.com/api
```

Do **not** hardcode localhost URLs.

GitHub Repository Structure

```
bike-rental-system/

backend/

frontend/

docs/
```

```
README.md

PROJECT_PLAN.md

SETUP_GUIDE.md
```

Git Branch Strategy

```
main

develop

feature/auth

feature/bikes

feature/rental

feature/dashboard

feature/mobile-ui

feature/deployment
```

Build Commands

Backend

```
mvn clean install

mvn spring-boot:run
```

Frontend

Open

```
index.html
```

or serve using

```
Live Server
```

Deployment Checklist

Backend

- Java Version = 21
- Build Successful
- REST APIs Tested
- Database Connected
- Environment Variables Configured
- Deploy to Render or Railway

Frontend

- Responsive Layout
- Mobile Tested
- API URL Updated
- Static Files Optimized
- Deploy to GitHub Pages
- Deploy to Vercel

Team Development Rules

- Pull latest changes before starting work.
- Work only on your assigned feature branch.
- Commit frequently with descriptive commit messages.
- Open Pull Requests into `develop`.
- Resolve merge conflicts before merging.
- Merge into `main` only after successful testing.

Testing Checklist

- User Registration
- Login
- Search Bikes
- Rent Bike
- Return Bike
- QR Code Generation
- Mobile Responsiveness
- Admin Dashboard
- Analytics Charts
- REST API Responses
- Deployment Verification

Future Improvements

- Progressive Web App (PWA)
- Push Notifications
- Google Maps Integration
- Payment Gateway
- AI-Based Bike Recommendation
- Real-Time Ride Tracking

Final Deliverables

- Responsive Mobile-Friendly Web Application
- Java Spring Boot Backend
- REST API Integration
- Admin Analytics Dashboard
- GitHub Repository
- Frontend Deployed on GitHub Pages
- Frontend Deployed on Vercel
- Backend Deployed on Render or Railway
- Complete Documentation
- Working Live Demonstration